AP PRECALCULUS

Unit 1 • Lesson 1.8 • Notes (filled in)

Topic 1.8 • Rational Functions and Zeros

Notes (filled in)

Learning goals

- Find the real zeros of a rational function, checking the domain.
- Solve rational inequalities with a sign chart.

Vocabulary. zero, domain, excluded value, sign chart, test value, interval notation.

1. Zeros

Key idea

The real zeros of $r = \dfrac{p}{q}$ are the real zeros of __p__ that are __in the domain of r__ (not zeros of q).
Always __factor__ first.

Example 1

Find the zeros of $r(x) = \dfrac{x^2 - 3x}{x^2 - 9}$ and of $s(x) = \dfrac{x^2 - 4}{x + 1}$.

$r(x) = \frac{x(x-3)}{(x-3)(x+3)}$: numerator zeros $0, 3$, but $x = 3$ is excluded. Zero: $x = 0$ only.
s: zeros $x = 2$ and $x = -2$ (neither is excluded).

2. Sign charts

The sign of a rational function can change only at its __zeros__ and at __values excluded from its domain__.
Mark these on a number line, test __one value in each interval__, and read off the sign.

Example 2 • Solve $r(x) \le 0$

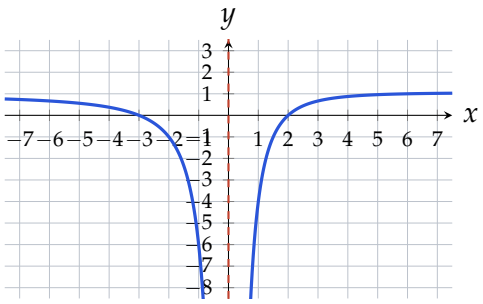
Solve $\dfrac{(x+1)(x-4)}{x-2} \le 0$.

Key values: $-1, 2, 4$.



Tests: $x = -2$: negative; $x = 0$: positive; $x = 3$: negative; $x = 5$: positive. Include zeros, exclude $x = 2$: $(-\infty, -1] \cup (2, 4]$.

Example 3 • From a graph



The graph of $r(x) = \frac{(x+3)(x-2)}{x^2}$ is shown. Solve $r(x) > 0$.

The graph is above the axis for $x < -3$ and $x > 2$: $(-\infty, -3) \cup (2, \infty)$.

Common mistakes

- Including excluded values in the solution of $\leq$ or $\geq$.
- Multiplying both sides of an inequality by an expression whose sign is unknown.

Check your understanding

1. Zeros of $\frac{x^2-1}{x+1}$: $x = 1$ only .

2. $\frac{x}{x-3} > 0$ on $(-\infty, 0) \cup (3, \infty)$.